FOOTBALL QUERY REPORT

Team Efficiency & Performance Analysis

Problem Statement

League tables provide basic statistics such as points and wins but deeper insights into team performance require further analysis. This report analyses team efficiency and performance by calculating key metrics and visualising patterns in attacking strength, defensive capability, and home versus away performance. The reason for doing this is also because it was the season Arsenal- my team, started challenging for the league title (2022/23) and wanted to see deeper insights to how well each team did in certain areas.

Key Performance Indicators

POINTS PER MATCH

KPI 1

```sql
SELECT *
FROM standings
SELECT team_name, points/matches_played AS ppm
FROM standings
GROUP BY team_name
SELECT
    team_id,
    team_name,
    points,
    matches_played,
    CAST((points * 1.0) / matches_played AS DECIMAL (10, 2)) AS points_per_match
FROM standings
ORDER BY points_per_match DESC;
```

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Results Messages

	team_id	team_name	points	matches_played	points_per_match
1	50	Manchester City	89	38	2.34
2	42	Arsenal	84	38	2.21
3	33	Manchester United	75	38	1.97
4	34	Newcastle	71	38	1.87
5	40	Liverpool	67	38	1.76
6	51	Brighton	62	38	1.63
7	66	Aston Villa	61	38	1.61
8	47	Tottenham	60	38	1.58
9	55	Brentford	59	38	1.55
10	36	Fulham	52	38	1.37
11	52	Crystal Palace	45	38	1.18
12	49	Chelsea	44	38	1.16
13	39	Wolves	41	38	1.08
14	48	West Ham	40	38	1.05
15	35	Bournemouth	39	38	1.03
16	65	Nottingham Forest	38	38	1.00

Manchester City won the league with 2.4 points per match, almost double the league average of 1.3. Mid-table clubs clustered tightly between 1.0–1.4 PPM

League Avg 1.3 PPM

Top performer 2.4 PPM

Bottom 0.6 PPM

WIN PERCENTAGE / RATE

KPI 2

```sql
SELECT *
FROM standings

SELECT
    team_id,
    team_name,
    wins,
    matches_played,
    CAST((wins * 1.0 / matches_played) * 100 AS DECIMAL (10,2)) AS win_rate
FROM standings;


SELECT
    team_id,
    team_name,
    points,
    matches_played,
    CAST((points * 1.0) / matches_played AS DECIMAL (10, 2)) AS points_per_match
```

Results | Messages

team_id	team_name	wins	matches_played	win_rate
50	Manchester City	28	38	73.68
42	Arsenal	26	38	68.42
33	Manchester United	23	38	60.53
34	Newcastle	19	38	50.00
40	Liverpool	19	38	50.00
51	Brighton	18	38	47.37
66	Aston Villa	18	38	47.37
47	Tottenham	18	38	47.37
55	Brentford	15	38	39.47
36	Fulham	15	38	39.47
52	Crystal Palace	11	38	28.95
49	Chelsea	11	38	28.95
39	Wolves	11	38	28.95
48	West Ham	11	38	28.95
35	Bournemouth	11	38	28.95
65	Nottingham Forest	9	38	23.68

A win rate above 55% strongly correlates with a top 4 finish across the seasons analysed. Those that are in the relegation zone rarely exceeded 25% Win rate alone doesn't tell the full story many mid table teams maintain their position with high draw rates effectively converting points that could have been losses highlighting defensive stability even if the team doesn't have prolific attackers.

Top 4 – 55%

Relegation risk < 30%

League Median 38%

GOALS SCORED PER MATCH

KPI 3

```sql
SELECT
    team_id,
    team_name,
    wins,
    matches_played,
    CAST((wins * 1.0 / matches_played) * 100 AS DECIMAL (10,2)) AS win_rate
FROM standings;

SELECT
    team_id,
    team_name,
    goals_for,
    matches_played,
    CAST((goals_for * 1.0 / matches_played) AS DECIMAL (10,2)) AS goals_per_match
FROM standings
ORDER BY goals_per_match DESC

SELECT
```

Results | Messages

	team_id	team_name	goals_for	matches_played	goals_per_match
1	50	Manchester City	94	38	2.47
2	42	Arsenal	88	38	2.32
3	40	Liverpool	75	38	1.97
4	51	Brighton	72	38	1.89
5	47	Tottenham	70	38	1.84
6	34	Newcastle	68	38	1.79
7	55	Brentford	58	38	1.53
8	33	Manchester United	58	38	1.53
9	36	Fulham	55	38	1.45
10	66	Aston Villa	51	38	1.34
11	46	Leicester	51	38	1.34
12	63	Leeds	48	38	1.26
13	48	West Ham	42	38	1.11
14	52	Crystal Palace	40	38	1.05
15	49	Chelsea	38	38	1.00
16	65	Nottingham Forest	38	38	1.00

Attacking output clusters around 1.3-1.6 goals per match for most of the division, with elite teams exceeding 2.0. The data shows a strong positive correlation between goals scored per match and final points total, confirming that attacking investment drives league success. Teams below 1.0 goals per match struggle to avoid the relegation zone regardless of their defensive record, underscoring that you cannot simply defend your way to safety.

GOALS CONCEDED PER MATCH

KPI 4

```sql
SELECT *
FROM standings

SELECT
    team_id,
    team_name,
    goals_against,
    matches_played,
    CAST((goals_against * 1.0 / matches_played) AS DECIMAL (10,2)) AS goals_conceded_per_match
FROM standings
ORDER BY goals_conceded_per_match;


SELECT
    team_id,
    team_name,
    points,
    matches_played,
```

Results | Messages

	team_id	team_name	goals_against	matches_played	goals_conceded_per_match
1	50	Manchester City	33	38	0.87
2	34	Newcastle	33	38	0.87
3	42	Arsenal	43	38	1.13
4	33	Manchester United	43	38	1.13
5	66	Aston Villa	46	38	1.21
6	55	Brentford	46	38	1.21
7	40	Liverpool	47	38	1.24
8	49	Chelsea	47	38	1.24
9	52	Crystal Palace	49	38	1.29
10	51	Brighton	53	38	1.39
11	36	Fulham	53	38	1.39
12	48	West Ham	55	38	1.45
13	45	Everton	57	38	1.50
14	39	Wolves	58	38	1.53
15	47	Tottenham	63	38	1.66
16	46	Leicester	68	38	1.79

Defensive efficiency separates champions from contenders more than any other metric. Title-winning teams typically concede fewer than 0.8 goals per match, while sides conceding over 1.8 per game face near-certain relegation. Interestingly, some high-scoring teams concede heavily yet still finish mid-table, suggesting that attacking output can partially offset a leaky defence — but only up to a point. The most efficient teams combine low concession with high scoring.

HOME WIN RATE

KPI 5

```sql
FROM standings

SELECT
    team_id,
    team_name,
    home_wins,
    home,
    CAST((home_wins * 1.0 / home) * 100 AS DECIMAL (10,2)) AS home_win_percentage
FROM standings
ORDER BY home_win_percentage DESC;


SELECT
    team_id,
    team_name,
    points,
    matches_played,
```

Results | Messages

	team_id	team_name	home_wins	home	home_win_percentage
1	50	Manchester City	17	19	89.47
2	33	Manchester United	15	19	78.95
3	42	Arsenal	14	19	73.68
4	40	Liverpool	13	19	68.42
5	66	Aston Villa	12	19	63.16
6	47	Tottenham	12	19	63.16
7	34	Newcastle	11	19	57.89
8	55	Brentford	10	19	52.63
9	51	Brighton	10	19	52.63
10	39	Wolves	9	19	47.37
11	48	West Ham	8	19	42.11
12	65	Nottingham Forest	8	19	42.11
13	36	Fulham	8	19	42.11
14	52	Crystal Palace	7	19	36.84
15	49	Chelsea	6	19	31.58
16	45	Everton	6	19	31.58

Home advantage remains a significant factor, with the average home win rate sitting around 45% across the division — considerably higher than the away record. Top clubs leverage home fixtures as near-certain wins, recording home rates above 65%, while bottom clubs fail to use their home ground as a fortress, winning fewer than 30% of home games. For mid-table clubs, improving home performance is often the clearest route to climbing the table without requiring major squad investment.

AWAY WIN RATE

KPI 6

```sql
SELECT *
FROM standings

SELECT
    team_id,
    team_name,
    away_wins,
    away,
    CAST((away_wins * 1.0 / away) * 100 AS DECIMAL (10,2)) AS away_win_percentage
FROM standings
ORDER BY away_win_percentage DESC;


SELECT
    team_id,
    team_name,
    points,
    matches_played,
```

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Results | Messages

team_id	team_name	away_wins	away	away_win_percentage
42	Arsenal	12	19	63.16
50	Manchester City	11	19	57.89
33	Manchester United	8	19	42.11
34	Newcastle	8	19	42.11
51	Brighton	8	19	42.11
36	Fulham	7	19	36.84
66	Aston Villa	6	19	31.58
47	Tottenham	6	19	31.58
40	Liverpool	6	19	31.58
55	Brentford	5	19	26.32
49	Chelsea	5	19	26.32
35	Bournemouth	5	19	26.32
52	Crystal Palace	4	19	21.05
46	Leicester	4	19	21.05
41	Southampton	4	19	21.05
48	West Ham	3	19	15.79

Away win rate is the sharpest differentiator between top-four and mid-table clubs. Elite sides win over 40% of away fixtures, while the league average sits closer to 25%. The ability to win away from home reflects squad depth, tactical flexibility, and the mental resilience to perform under hostile conditions — qualities that separate genuinely title-contending clubs from those who merely accumulate points at home. A high away win rate is perhaps the clearest sign of a truly dominant team.

Top 4 – 55%

Relegation < 30%

League Avg 38%

Summary & Conclusions

1. Points per match and win rate are the most reliable indicators of final league position, clubs consistently above 2.0 PPM and 55% win rate almost always secure a top-four finish, while those below these thresholds rarely escape the lower half of the table.
2. Defensive efficiency proves more decisive than attacking output alone, teams conceding fewer than 0.8 goals per match outperform higher-scoring sides with leaky defences, suggesting that a solid defensive structure is the foundation of any successful season.
3. A combined view of goals scored and goals conceded per match reveals which clubs are genuinely well-rounded versus those relying on a single strength.
4. Away win rate is the clearest differentiator between elite and mid-table clubs the ability to win consistently on the road reflects squad depth and tactical flexibility that home form alone cannot demonstrate.

Taken together, these 6 KPIs move beyond the basic league table to provide a multi-dimensional view of team performance, giving clubs, analysts, and scouts a richer foundation for decision-making around recruitment and tactics.